A LEFT DUAL IN THE MONOIDAL CATEGORY OF $\MATHCAL{O}$-MODULES IS LOCALLY A SUMMAND OF A FINITE FREE MODULE; THE EVALUATION FORMULA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate and re-verified against the pinned source (sm-twins-20261014.json span sanity: all OK).

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found hard occupation only from the 47 pre-existing lane-s11 S11-sites-mod-* cards (same pinned snapshot, delivered) and cross-chapter citation noise, 0 other-lane hits for sites-modules.tex (sm-occupied.json, 2026-10-14); the RESUME-pool dispatch unlocks the remaining face.

Source reference. Original reference 0FFP: sources/stacks-chow/tags.txt, source lines 19294–19294. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0FFQ: sources/stacks-chow/tags.txt, source lines 19295–19295. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0FFR: sources/stacks-chow/tags.txt, source lines 19296–19296. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0FNY: sources/stacks-chow/tags.txt, source lines 19583–19583. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{F}$ be a $\mathcal{O}$-module. Let $\mathcal{G}, \eta, \epsilon$ be a left dual of $\mathcal{F}$ in the monoidal category of $\mathcal{O}$-modules, see Categories, Definition 0FFP. Then*

1. *for every object U of $\mathcal{C}$ there exists a covering $\{U_i \to U\}$ such that $\mathcal{F}|_{U_i}$ is a direct summand of a finite free $\mathcal{O}|_{U_i}$-module,*
2. *the map $e : \mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{O}) \to \mathcal{G}$ sending a local section λ to $(\lambda \otimes 1)(\eta)$ is an isomorphism,*
3. *we have $\epsilon(f, g) = e^{-1}(g)(f)$ for local sections f and g of $\mathcal{F}$ and $\mathcal{G}$.*

Proof. The assumptions mean that

$$\mathcal{F} \xrightarrow{\eta \otimes 1} \mathcal{F} \otimes_{\mathcal{O}} \mathcal{G} \otimes_{\mathcal{O}} \mathcal{F} \xrightarrow{1 \otimes \epsilon} \mathcal{F} \quad \text{and} \quad \mathcal{G} \xrightarrow{1 \otimes \eta} \mathcal{G} \otimes_{\mathcal{O}} \mathcal{F} \otimes_{\mathcal{O}} \mathcal{G} \xrightarrow{\epsilon \otimes 1} \mathcal{G}$$

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are the identity map. Let U be an object of $\mathcal{C}$. After replacing U by the members of a covering of U, we can find a finite number of sections $f_1, \ldots, f_n$ and $g_1, \ldots, g_n$ of $\mathcal{F}$ and $\mathcal{G}$ over U such that $\eta(1) = \sum f_i g_i$. Denote

$$\mathcal{O}_U^{\oplus n} \to \mathcal{F}|_U$$

the map sending the ith basis vector to f_i. Then we can factor the map $\eta|_U$ over a map $\tilde{\eta} : \mathcal{O}_U \to \mathcal{O}_U^{\oplus n} \otimes_{\mathcal{O}_U} \mathcal{G}|_U$. We obtain a commutative diagram

$$\begin{array}{ccccc} \mathcal{F}|_U & \xrightarrow{\eta\otimes 1} & \mathcal{F}|_U \otimes \mathcal{G}|_U \otimes \mathcal{F}|_U & \xrightarrow{1\otimes\epsilon} & \mathcal{F}|_U \\ & \searrow^{\tilde{\eta}\otimes 1} & \uparrow & & \uparrow \\ & & \mathcal{O}_U^{\oplus n} \otimes \mathcal{G}|_U \otimes \mathcal{F}|_U & \xrightarrow{1\otimes\epsilon} & \mathcal{O}_U^{\oplus n} \end{array}$$

This shows that the identity on $\mathcal{F}|_U$ factors through a finite free $\mathcal{O}_U$-module. This proves (1). Part (2) follows from Categories, Lemma 0FFQ and its proof. Part (3) follows from the first equality of the proof. You can also deduce (2) and (3) from the uniqueness of left duals (Categories, Remark 0FFR) and the construction of the left dual in Example 0FNY. □